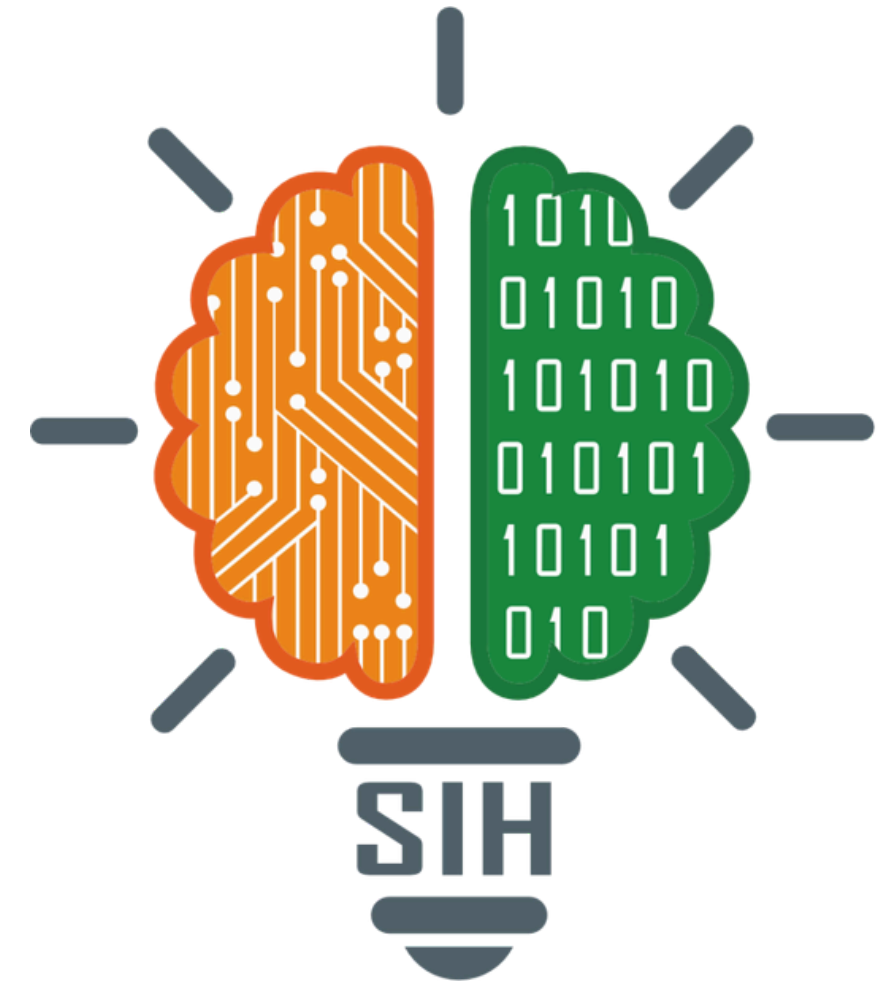


SMART INDIA HACKATHON 2026

- **Problem Statement ID** – 26043
- **Problem Statement Title** – A digital platform to crowdsource societal challenges and facilitate collaborative problem solving through universities and industry partnerships
- **Theme** – Smart Education
- **PS Category** – Software
- **Team ID** –
- **Team Name** – BilluSena



JanSetu is a bilingual societal innovation **collaboration portal** that **transforms community-reported challenges** in Jharkhand into reviewed, **university-led and industry-supported projects** with measurable outcomes.

Key Capabilities

→ Citizen Challenge Intake

Submit issues with evidence, location and supporting documents.

→ AI-Assisted Analysis

Suggests domain, priority, duplicates and university matches.

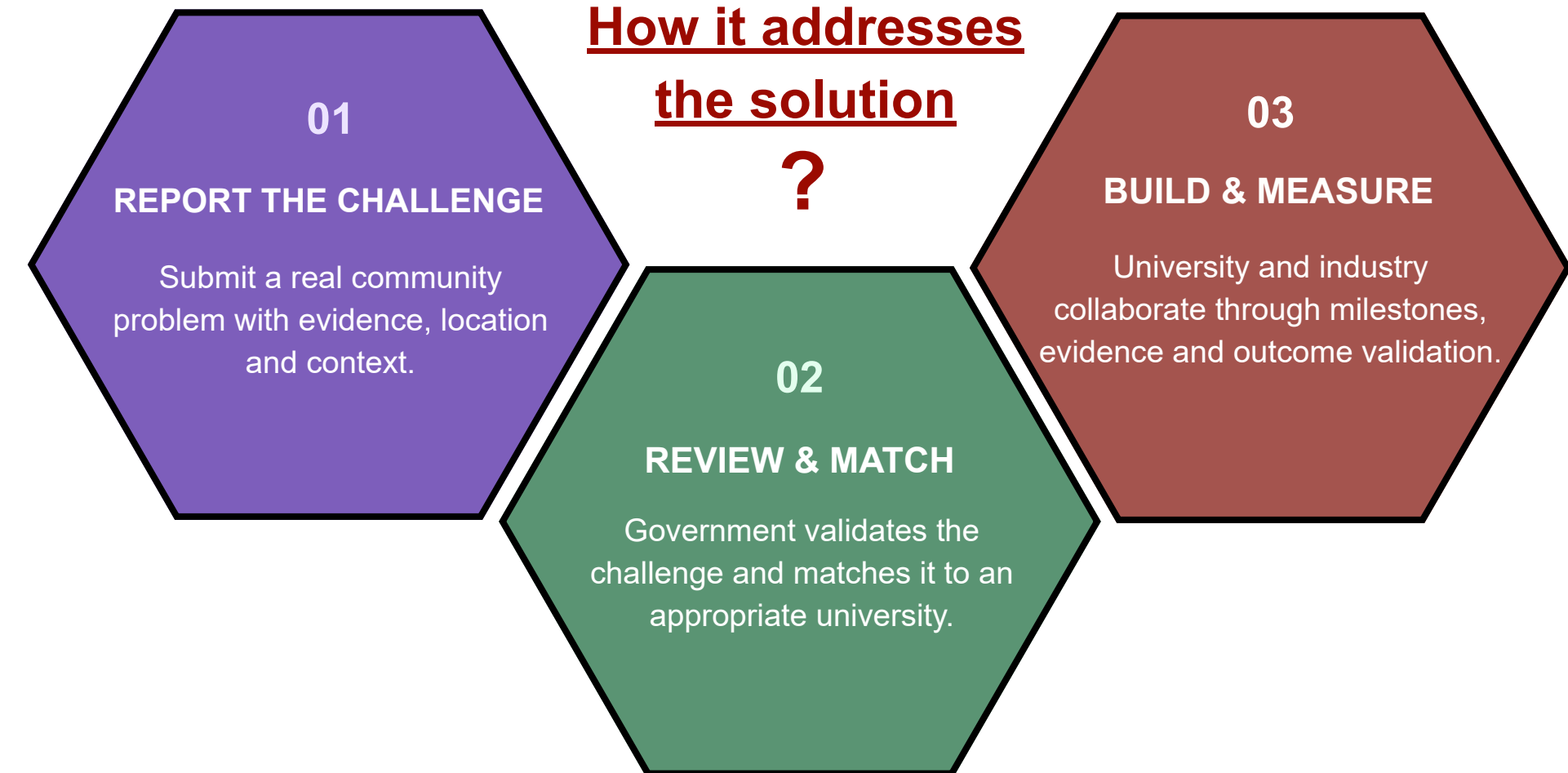
→ Government Review & Allocation

Validate challenges, refine summaries and assign universities.

→ University-Industry Collaboration

Build solutions through teams, mentorship, funding, pilots and technology transfer.

How it addresses the solution



Innovation and Uniqueness

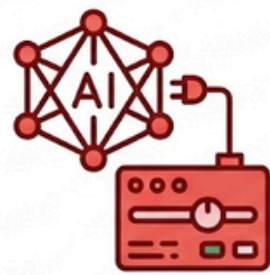
Bilingual by Design



Evidence-Backed Workflow

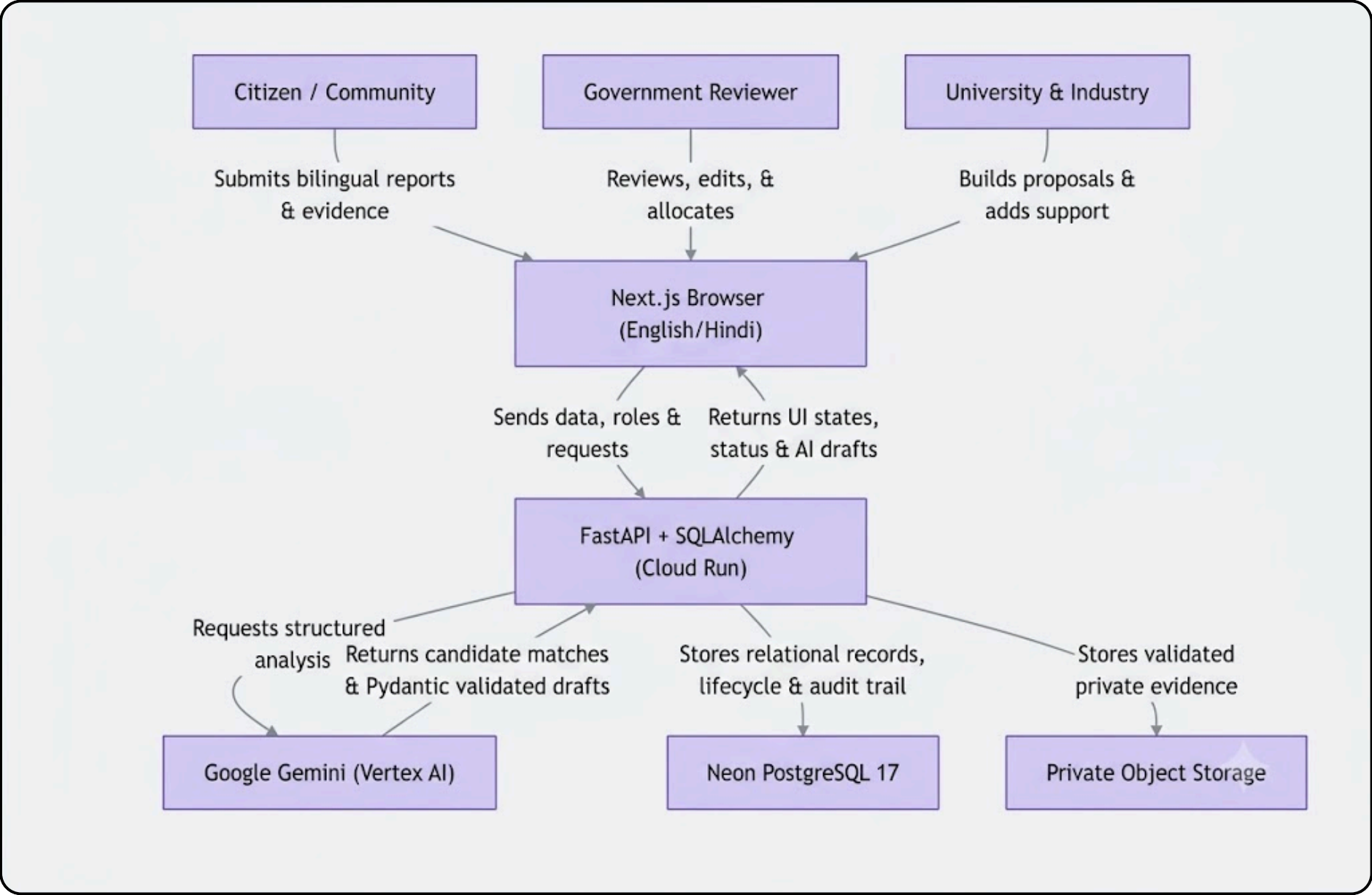


Human-Controlled AI

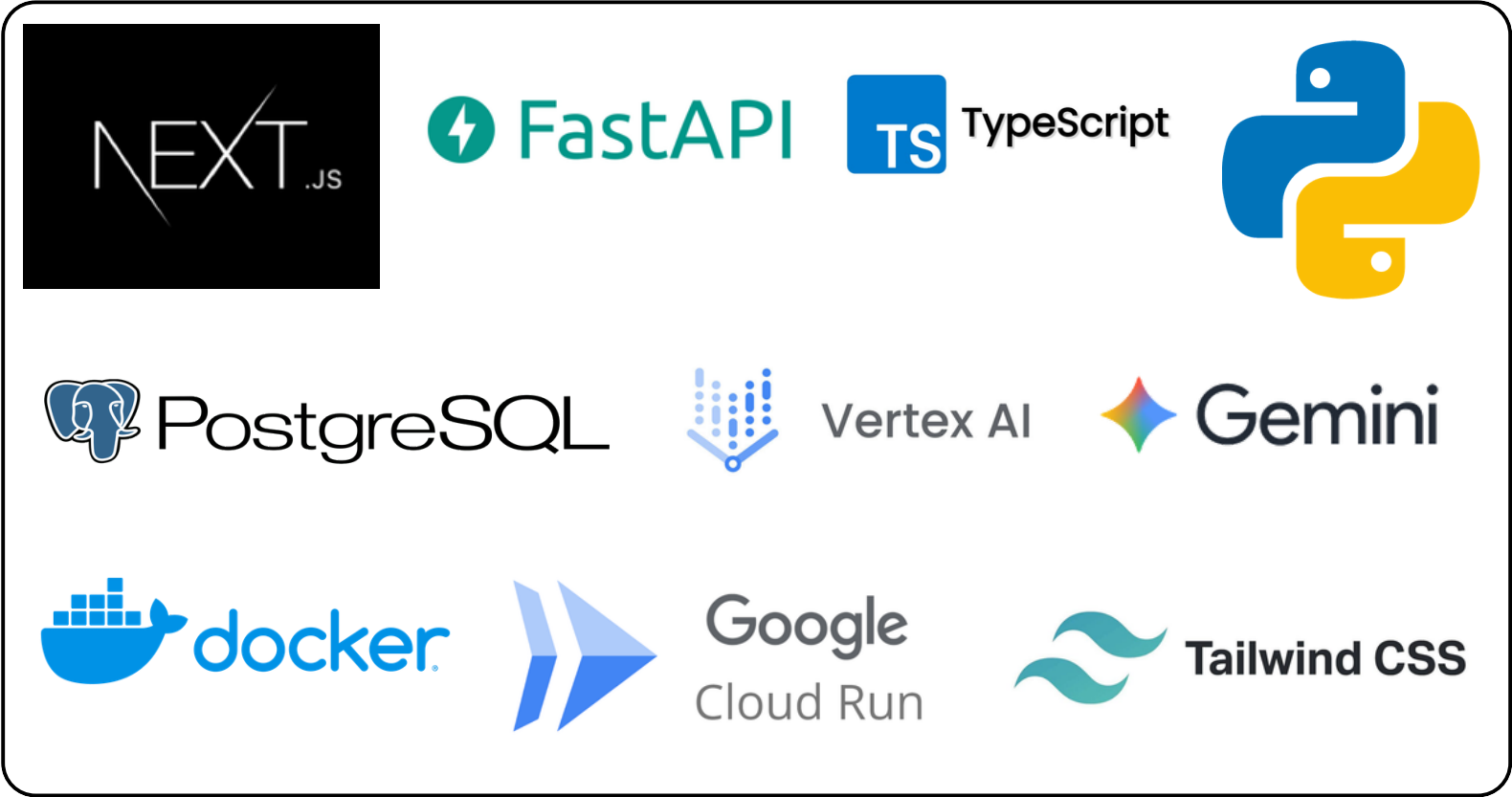


- **Bilingual by Design:** English + Hindi public fields, with human-reviewed bilingual summaries while preserving the original submission.
- **Human-Controlled AI:** Gemini assists with classification, prioritisation, duplicate detection and matching — but never publishes, assigns, approves, rejects or resolves a challenge.
- **End-to-End Lifecycle:** One connected workflow from challenge **intake** → **review** → **university allocation** → **proposal** → **partnership** → **milestones** → **outcome validation**.
- **Evidence + Accountability:** Private evidence, role-based access, activity history and government-validated outcomes create a transparent and **auditable innovation pipeline**.

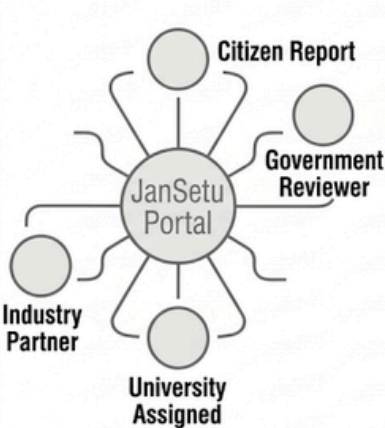
WORKFLOW



TECH STACK

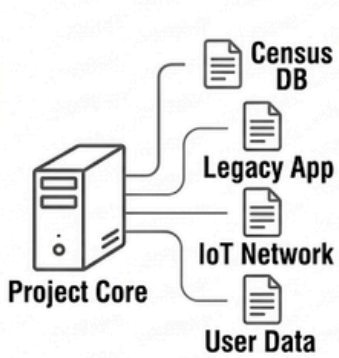


01 JanSetu Innovation Workflow



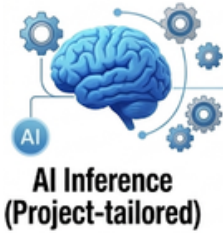
- Challenge Intake:** Multilingual report with private evidence.
- Government Review:** Triage, classification and duplication check.
- Project Assignment:** Direct university allocation based on needs.
- Validated Outcomes:** Accountable impact with private audits.

02 Evidence-First Verification Hub



- Cross-check Profiles:** Authenticate users from trusted sources.
- Secure Workflows:** Isolate core system processes for protection.
- Streamline Data:** Apply scalable data processing models.
- Event Logs:** Maintain immutable system event logs.

03 Conflict-Aware State & Decision



- Authentic Records:** Validated source history for all incoming data.
- Formula Transparency:** Calculation methodology is clear and publicly verifiable.
- System Checks:** Continuous, real-time monitoring of operational integrity.
- Human Oversight:** Operators retain all critical decision approval.

Potential Challenges & Risks

AI Suggestion Risks



Transient errors cause structured draft invalidation.

Complex Match Recall



Shortlist limit may cause missing potential best university matches.

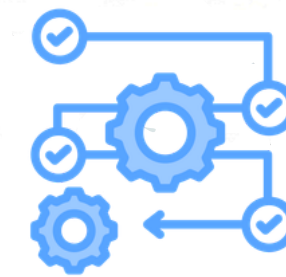
Operational Scaling



Onboarding complexity and storage needs add system complexity.

Strategies to Overcome

Advanced Context & Intermediate Checks



Integrate domain-specific context and pre-validate AI outputs.

Multi-Stage Recall & Dynamic Shortlisting



Use dynamic limits and secondary recall for a complete potential match pool.

Resilient Job Queues & Safe State Caching



Employ resilient job processing queues and state caching for system stability.

Analysis of Feasibility

Technical

Built on a mature open-source stack (Next.js, FastAPI, PostgreSQL). Enforces access control via REST endpoints and uses schema-validated AI output to prevent invalid categorizations.

Economic

Cost-effective through open-source technologies and usage-based scaling via Cloud Run and Vercel. Reduces manual triage workloads while leaving all final decisions to authorized human officials.

Operational

Integrates naturally by providing separate, dedicated workspaces for citizens, government, universities, and industry. Built-in notifications handle coordination without needing complex real-time external services.

Potential Impact On Target Audience:

Citizens & Communities:

Simple bilingual reporting of local challenges with evidence, progress visibility and feedback.

Government Departments:

Centralized review, transparent university allocation, workflow monitoring and validated impact tracking.

Universities & Industry:

Real-world problems for universities; mentorship, funding, prototypes and pilots for industry.

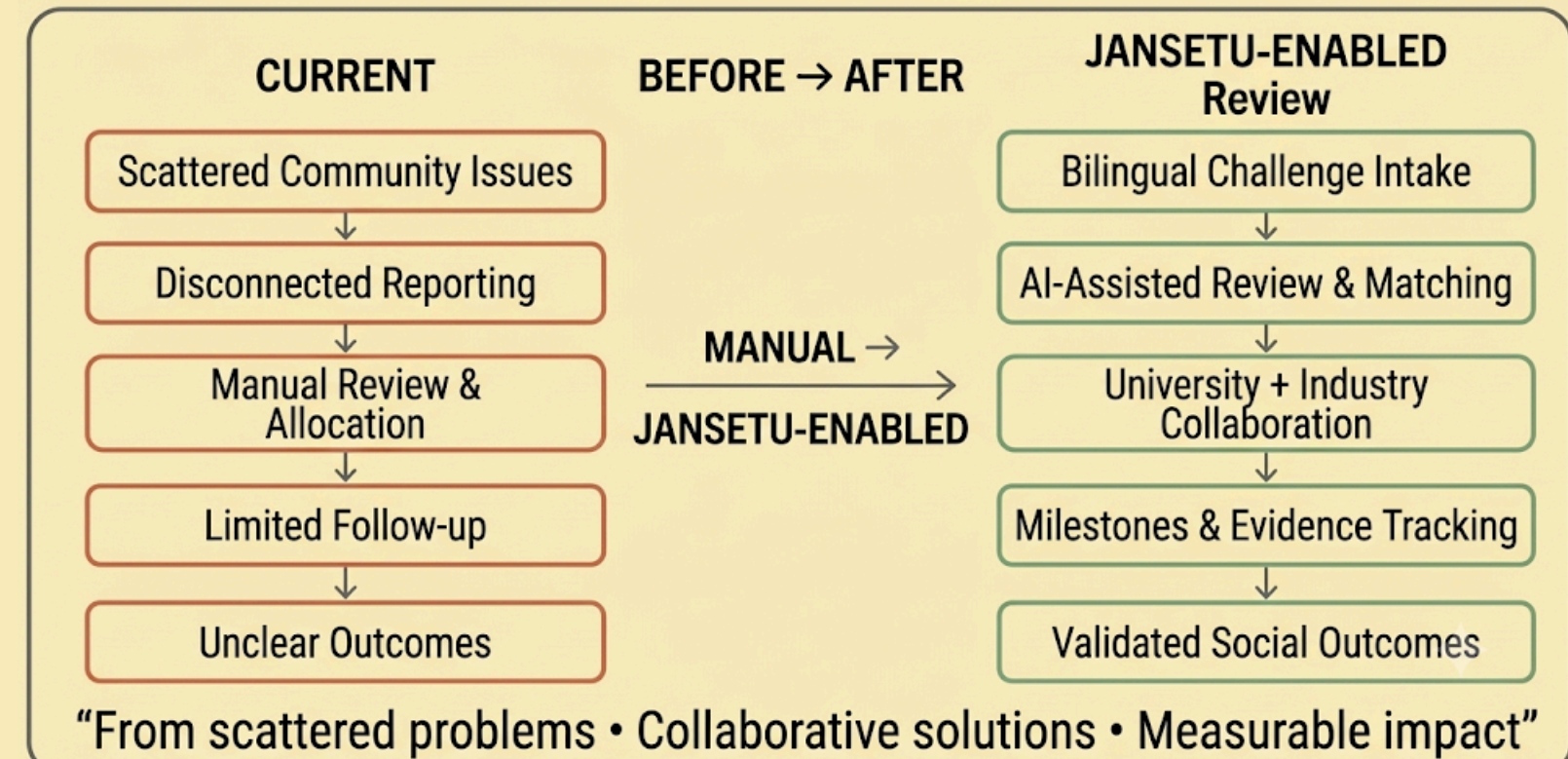
Key Metrics:

- Challenges by district & domain
- Active universities & projects
- Partnerships & funding commitments
- Validated outcomes & beneficiaries

Benefits Of The Solution:

- **Faster Problem Resolution:** Connects challenges to the right university and supporting partners.
- **Better Collaboration:** One platform for citizens, government, universities and industry.
- **Transparency & Accountability:** Reviews, milestones, evidence and outcomes remain traceable.
- **Bilingual & Inclusive:** English/Hindi reporting and communication.
- **Measurable Social Impact:** Tracks beneficiaries, results, testing evidence, patents and startups.

OPERATIONAL IMPACT



RESEARCH AND REFERENCES

Project Links:

Live Demo: <https://jansetu-jharkhand.vercel.app/>

Frontend: jansetu-jharkhand.vercel.app

Backend: jansetu-api-595351864949.us-east4.run.app/api/v1/health

Demo Video:

<https://drive.google.com/drive/folders/1-sVTEnGEMpwg4abO0D2a-73zSsLiS5oR?usp=sharing>

Research Papers:

Smart Civic Issue Reporting System (Vikas Path):

https://www.researchgate.net/publication/359118551_Smart_Civic_Issue_Reporting_System

IN-PART / Connect — University-Industry Matchmaking: <https://in-part.com/blog/university-industry-matchmaking/>

NLP-Based Grievance Management System with Code-Mixer Speech Analysis:

<https://papers.ssrn.com/sol3/Delivery.cfm/6369498.pdf?abstractid=6369498&mirid=1>

AI and Technical References:

Gemini Structured Output: <https://ai.google.dev/gemini-api/docs/structured-output>

Vertex AI — ADC Quickstart: <https://cloud.google.com/vertex-ai/generative-ai/docs/start/quickstart>

FastAPI: <https://fastapi.tiangolo.com/>

Neon: <https://neon.tech/docs>

Google Cloud Run: <https://cloud.google.com/run/docs>